



An introductory review of gravitational physics

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Gravitational physics is a field that centers around analyzing the physical implications and theoretical consequences of Einstein's theory of general relativity. For decades, gravitational physics was a niche field restricted to mostly theoretical conjecture and mathematical investigations. However, in the past few years, the detection of significant gravitational phenomena has caused gravitational physics to grow into a high-interest field with a multitude of subfields and applications in both theoretical and experimental realms. With the ability to now utilize data to test long-standing theories and motivate the development of new ones, as well as merge the study of gravitational physics with other areas of cosmology, it is clear we are entering a new age of gravity. As such, it is worthwhile to provide an accessible introduction to the field of gravitational physics to give insight into its rising relevance. This paper serves as a preliminary review of gravitational physics, beginning with the basic concept and ideas of general relativity and discussing some of the more prominent subfields of gravitational physics, such as alternate gravity theory, gravitational wave astronomy, and quantum gravity. We address the motivations behind alternative and modified theories of gravity and explain some of the most prevalent gravity theories, $f(R)$ gravity and Moffat's Scalar-Tensor-Vector gravity. We also discuss the existence of gravitational waves and some of the exciting prospects posed by gravitational waves astronomy. In addition, we examine the recent detection of a gravitational wave background by the North American Nanohertz Observatory for Gravitational Waves and the computation of the Hubble constant through gravitational wave data. We conclude by briefly assessing string theory and loop quantum gravity, the two most prominent theories in quantum gravity.

I. INTRODUCTION

The field of gravitational physics, which was once a small, relatively niche field chiefly focused on purely mathematical explorations of gravitational theory, has since grown into a high-interest field with theoretical and experimental applications in many areas of astrophysics and cosmology, as well as quantum physics. Some would even consider the theoretical questions and experimental challenges posed in the field of gravitational physics to be some of the most intriguing and complex in contemporary physics, especially in regards to what is known as *quantum gravity* [1]. As such, this begs the question: What does the field of gravitational physics entail?

The core of gravitational physics centers around the theory of general relativity. Published in 1915, this theory expanded upon Einstein's previous theory of special relativity [2] and Newton's law of universal gravitation by establishing a connection between the nature of space-time and gravitational forces [3]. Notably, Einstein determined that the presence of mass in four-dimensional spacetime induces spacetime curvature, an occurrence which determines the motion of objects and manifests as gravity. This concept dictates the essence of the field of gravitational physics and has been revolutionary in motivating our understandings of astrophysical phenomena. Most of our modern presumptions of gravity orient around this theory.

As such, gravitational physics research involves exper-

imental and continued mathematical investigation into the theory of general relativity, as well as tests of this theory's relationship to quantum mechanics. Many subfields also explore the consequences of this theory on physical phenomena, such as the detection and computation of gravitational waves and the conceptualization of dark matter and energy. Gravitational physics additionally has relevance in understanding and motivating galaxy and cluster formation, black hole dynamics, as well as many other astrophysical and cosmological events.

This paper will cover some of the more prominent areas of study of gravitational physics, including alternative and modified theories of gravity, gravitational wave theory and experiment, and quantum gravity. We will begin each section by covering the physical motivations and concepts as well as some of the historical precedence behind each field. We will close each section by exploring some examples and references of current research and investigation, in addition to potential areas of future scientific development. We will conclude the review with a summary of these subfields as well as the field as a whole.

II. ALTERNATIVE AND MODIFIED GRAVITY

It is exceedingly rare to encounter a theory within the field of physics that was successful upon its initial iteration and did not require continued significant modifications by multiple physicists. In this regard, the theory

of general relativity is rather unique. Thus far, it has remained steadfast against multiple experimental tests, including examinations on the scale of the solar system and through gravitational wave astronomy (see section III) [4]. However, there are still ongoing investigations into general relativity's validity in the strong-field regime, that being the regions of spacetime that are intensely curved and where gravitational effects are particularly strong. Furthermore, it is critical to note that many popular physics investigations and theories, such as the concept of dark matter and energy as well as the notion of an accelerating universe, are a direct result of applications of general relativity to observed cosmological data. Dark matter and energy alone are still disputed concepts within the field of gravitational physics, being regarded as a rather exotic notion by some [5]. This is in addition to the fact that dark matter and dark energy have yet to be directly tested for and definitively proven to exist in a lab setting [6].

As such, the testing of general relativity is of great relevance and vital to our understanding of the physical universe. It is additionally worthwhile to investigate alternatives to this theory, ones that could possibly explain various physical phenomena through differing means and/or conceivably justify the disconnects between general relativity and quantum mechanical theory (see section IV). Such attempts to devise alternative theories of gravity have been ongoing since Einstein's first publication on general relativity (see [6–8]). Most have, however, been disproven [9]. Regardless, there is still an active plethora of modern theories which are explored and investigated to this day, some of which we will cover in the following section.

A. Modern Theories of Gravity

Most prevailing modern theories of gravity function as modifications to general relativity rather than full alternatives to the theory itself. Many of these modifications follow the reasoning of the *Lovelock theorem* [10]. The Lovelock theorem is used to rationalize modifications to general relativity accomplished via modifications to the Einstein field equations, most commonly expressed as a tensor equation which relates the energy, momentum, and stress of objects within a region of spacetime to that of the curvature of local spacetime via the *stress-energy tensor* and the *Einstein tensor* respectively [3, 11, 12]. In common notation, the field equations are described by eq. (1), where $G_{\mu\nu}$ denotes the Einstein tensor, Λ denotes the cosmological constant (see section 11 of [11]), $g_{\mu\nu}$ denotes the metric tensor, κ denotes the Einstein gravitational constant, and $T_{\mu\nu}$ denotes the stress-energy tensor [3, 11]

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}. \quad (1)$$

As such, the theorem implies that there are five extant

methods to modify the field equations:

1. Add new fields in addition to or in replacement of the metric tensor.
2. Abandon the locality of the equation.
3. Work in a spacetime with a differing number of dimensions.
4. Add additional derivatives to the metric beyond the second-order.
5. What may be summarized as emergence from the field equations, which generally involves by-passing the condition of covariance in Lovelock's theorem [12–14].

In modern theories of modified gravity, this has translated to four common categorizations of modification: Theories beyond the second order relating to the fourth method, extra field content theories which relate to the first method, higher dimensional theories relating to the third method, and non-minimal coupling theories which relate to the fifth method¹ [13]. There are many examples of contemporary theories which fall under these classifications, but one of the more common ones is $f(R)$ gravity, a prominent family of modified gravity theories which are beyond second order [6, 13, 16].

1. $f(R)$ Gravity

$f(R)$ gravity may be described as a type of framework for generalized modifications of general relativity, first proposed by Buchdahl in 1970 [17]. More specifically, $f(R)$ gravity theories may be described as making a generalization of the Lagrangian in the Einstein-Hilbert action (see section 1.C of [6]) by defining a function f of the Ricci scalar R , which is a measure of curvature of a Riemannian manifold (see [18]), and using it to modify the action. Hence, $f(R)$ gravity. The simplest of these modifications comes in the form of defining $f(R)$ to be R itself. This case is equivalent to that of the base theory of general relativity.

To explain this in previously defined terms, the Einstein-Hilbert action is the action (a scalar describing the dynamics of a physical system) that yields the Einstein field equations, as shown in eq. (1). Modifications to this action will translate to modifications to the field equations. Specifically, this modification results in the relation shown in eq. (2), where $T_{\mu\nu}^{\text{eff}}$ is the effective stress-energy tensor [6, 12, 16]

¹ Modified theories of non-local gravity exist, but are simply not as common as the other methodologies of modification. An example of a non-local theory of gravity is found in [15].

$$G_{\mu\nu} = \frac{\kappa}{f'(R)}(T_{\mu\nu} + T_{\mu\nu}^{\text{eff}}). \quad (2)$$

When $f(R) = R$, eq. (2) becomes eq. (1). Further reading on these modifications and their consequences may be found in [6, 16].

The reasoning for making this kind of alteration to general relativity is that it justifies the accelerated expansion of the universe while eliminating the need for hypotheses of dark energy and dark matter [6]. However, as previously established, it is not the only theory that attempts to reconstruct gravity to eliminate the need for dark matter.

In spite of the prevalence of modified gravity theories, there also still exist fully alternate theories of gravity which find relevance today. These theories are usually constructed with the intent of fully replacing general relativity, rather than simply modifying it. One prominent example is Modified Newtonian Dynamics, also known as MOND, which was first proposed in 1983 and is built on the notion of modifying Newtonian laws of gravity to construct a new idea of gravitation [19]. However, there are many alternate theories of gravity which are still in use today, and many more which are still actively being devised. We will cover Moffat's Scalar-tensor-vector gravity (STVG), which also disregards dark matter while attempting to explain other gravitational phenomena. It is one of the more well-regarded alternate theories of gravitation [12, 20].

2. Moffat's Scalar-Tensor-Vector Gravity

STVG, first proposed in 2006 [20], is an extension of metric-skew-tensor gravity theory (MSTG) [21, 22]. Proposed in 2005, MSTG operates by employing a skew symmetric tensor field, $F_{\mu\nu\lambda}$, in addition to the Einstein tensor. This translates to the modified version of eq. (1) shown in eq. (3)

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}, \quad (3)$$

as well as an additional field equation for $F_{\mu\nu\lambda}$, which is defined as equation 10 in [21]. These equations require a variable version of Newton's constant G , as well as a varying coupling constant γ_c associated with the strength of the coupling of $F_{\mu\nu\lambda}$ to matter and a varying mass μ of $F_{\mu\nu\lambda}$. However, they allowed MSTG to justify the gravitational lensing of galaxy clusters without the need for dark matter and energy. Regardless, there was room for improvement within the framework of this theory, which was addressed in STVG.

STVG extended these principles and fixed some of their notable shortcomings. In particular, STVG made the significant adjustment of describing G , γ_c , and μ as dynamical scalar fields rather than dynamic variables

based on distance and time. This meant the theory was contained by a tensor, vector, and three scalar fields, with additional actions and a potential function to describe the vector field (see [5, 20]) and operated under a similar structure to that of MSTG. However, notably, STVG is able to better explain galaxy rotation curves, gravitational lensing, as well as other astrophysical phenomena compared to that of MSTG [20]. It also is in agreement with general relativity on smaller scales, which would justify previous experimental results regarding general relativity conducted within the solar system [4]. However, in light of gravitational waves having been successfully detected since STVG was first proposed, it remains to be seen whether STVG can justify the results of research seen in the following section.

III. GRAVITATIONAL WAVE THEORY AND EXPERIMENT

We previously established in the introduction that the existence of massive objects in the fabric of spacetime induces curvature. We also further established that the energy, mass, and stress of an object is related to the spacetime curvature through eq. (1). Based on these principles, it is not unreasonable to make the conjecture that massive accelerating bodies may induce rippling effects in spacetime². Considering the conception established in Einstein's theory of general relativity [3, 23], these rippling effects would presumably propagate at the speed of light, in some ways similar to electromagnetic radiation. This is known as *gravitational radiation*, or *gravitational waves*. Einstein himself foresaw the existence of these waves, and first described this phenomena as a result of his theory of general relativity in 1916 [3, 23]. However, Einstein was unsure that these waves would ever be detectable. This was due to the fact that the energy of a wave at a given point largely scales in relation to the mass and inversely to the distance [23]. As such, most sources of gravitational radiation near to Earth emit radiation of such small scale it is undetectable, and the greater sources of radiation would have lost significant energy by the time they reach Earth. This being the case, the only likely scenario in which a gravitational wave would be detected is if it is sourced by some of the most intense gravitational events in the universe, that being the collision of binary black holes and neutron stars [23].

Consequently, for many decades, there has been a great pursuit to attempt to detect gravitational waves from these sources through various experimental means. Outside of pursuing these detections in order to test the validity of general relativity, especially in the strong field

² This is the short explanation for how gravitational waves are induced, but is not exact. A full explanation for the physical and mathematical motivations behind gravitational radiation may be found in [23].

regime, the existence of these waves would result in an entirely new avenue for astronomy [23, 24]. It would provide a reliable method to investigate the properties of black holes and neutron stars, given that they usually are not detectable by visual means. Not to mention the many areas of cosmological research that could be explored using expanded black hole and neutron star data collected from gravitational waves. Fortunately, these research opportunities did not have to remain hypothetical as, after decades of experimental investigation, we finally found a successful detection in 2015.

Although it was originally constructed starting in 1992 and began searches in 1998 [25], it was not until in September of 2015 that the Laser Interferometer Gravitational-Wave Observatory (LIGO) [25] would make the first confirmed detection of gravitational waves from a binary black hole merger event [26]. This was a revolutionary discovery which started a whole new realm in research possibilities for gravitational physics. Not only that, but it would only take two more years before a binary neutron star inspiral was detected as well [27]. Since then, many more detections and observations have been made and there has been a plethora of research investigations into these events. In the following section, we will explore some of the more recent breakthroughs in research, particularly the measuring of the Hubble constant utilizing gravitational wave data and the novel detection of a gravitational wave background.

A. Recent Research Prospects

1. Measuring the Hubble Constant

In addition to being detected by LIGO, the event that caused the first binary black hole gravitational wave detection [26], referred to as GW170817, was also detected by optical astronomy facilities [28]. This additional observation allowed for the absolute distance to the source to be determined, thus the Hubble constant could be measured from these observations, giving a result of $70.0^{+12.0}_{-8.0} \text{ km s}^{-1} \text{ Mpc}^{-1}$ [28].

The Hubble constant is notorious for being one of the primary sources of uncertainty in contemporary cosmology, with multiple methods of measurement returning varying results for the expected value [29]. This poses a significant issue, as the Hubble constant is responsible for determining the rate of expansion of the Universe. As such, constraining the Hubble constant is one of the primary concerns of cosmology. The ability to potentially detect gravitational events and use them to derive a value for the Hubble constant could prove effective at helping characterize it, as it is distinct from prior methods of measurement. Although so far we have only been able to derive a value utilizing data from GW170817, as newer and more effective detectors are being constructed within the next two decades there is a greater chance that we will be able to better merge the data collected from

gravitational wave detectors and optical facilities.

2. NANOGrav Gravitational Wave Background

Perhaps one of the larger announcements within recent years, in July of 2023 the North American Nanohertz Observatory for Gravitational Waves (NANOGrav) announced their detection of a gravitational wave background in a 15 year data set of pulsar timing arrays [30]. This observation was of particular significance, as the gravitational waves detected through these pulsar timing arrays were of a much lower frequency than those detected by the likes of LIGO and other interferometers. The most likely sources for waves of this frequency is binary supermassive black hole mergers [30]. There is still much we have to learn about how supermassive black holes behave and form, and the detection of them through gravitational waves means we have the potential to investigate the properties of these illusive compact objects. Additionally, it has been theorized that many supermassive black holes may have been seeded in the very early years of the universe [31], meaning the characterization and study of supermassive black holes could potentially allow us to probe into the nature of the universe's formation.

Furthermore, a full description of the stochastic gravitational wave background through further exploration of NANOGrav data could give us the ability to potentially study the first second of the universe via the detection of primordial gravitational wave sources. Although it is too early after the initial detection for any solid research investigations to have been made yet, future research into using pulsar timing arrays for gravitational wave detections is proving to be promising.

IV. QUANTUM GRAVITY

As previously established in sections I and II, the currently accepted principles of general relativity and quantum mechanics are not congruent with one another. This was evident very early on in the history of the field, with Einstein himself pointing out in 1916 that quantum effects must play a role in the theory of general relativity. The disconnect between these two theories poses particular issue as there are many astrophysical domains where both gravitational and quantum mechanical effects cannot be overlooked, such as in the environment of black holes and internal structure of neutron stars [32] as well as the moments succeeding the Big Bang [33]. As such, the development of a theory which encapsulates both the principles of general relativity and the principles of quantum mechanics is critical and research on the subject is extensive. Thus far, there are two main categories which dominate the field of quantum gravity: Conceptions that fall under string theory and conceptions that fall under

loop quantum gravity. We will cover both of these groupings in the following section.

A. Theories of Quantum Gravity

1. String Theory

The central idea of string theory is that, rather than describing particles in a way that is point-like, they are described by one-dimensional objects known as *strings* [34]. Strings with different modes of oscillation will appear as particles with different charges and other distinguishing properties. Critically, one of the vibrational states of the string is associated with the graviton, a particle which carries gravitational force (in essence, the gravitational equivalent of a photon to light). This formulation comes at the expense of additional spacial dimensions in addition to the standard four-dimensional framework of spacetime [34].

Notably, the framework of string theory is mostly oriented around particle physics and thus has found many applications within that particular field [35]. For the field of gravitational physics, a more generalized framework that unifies it with the conceptions of general relativity was developed called superstring theory [36]. A more technical overview of this theory, which would be beyond the scope of this review, is provided in [37]. Although this theory has proved promising, it is still highly debated and does not always give a satisfactory definition or solution. Additionally, there are many odd consequences to the theory that have not fully been explained in a manner that is physically satisfying.

2. Loop Quantum Gravity

Loop Quantum Gravity (LQG) is one of the leading approaches to tackling quantum gravity, outside of string theory. Proposed in 1987, the basis of LQG is that the structure of spacetime may be described as a series of fine loops contained within a network. The connection of these loops are referred to as spin-connections, and the larger framework is referred to as the spin network [38]. This concept, on some level, means the spacetime metric itself becomes a quantum operator [39]. Although LQG is noted as being more rooted in fundamental physical concepts compared to string theory [39], there is little experimental corroboration of the theory and it has so far not been proven as a solid conjunction between or alternative to current quantum mechanical and general relativity theories. More detail on the technicalities

of LQG and quantum gravity can be found in [32, 36, 38].

V. CONCLUSIONS

Essentially existing on the boundary of two frontiers, that of the astrophysical and the quantum mechanical, gravitational physics is a vast field which poses some of the most challenging questions in contemporary physics. Centering around the theory of general relativity, gravitational physics explores ideas such as the dynamics of black holes and neutron stars, galaxy and cluster formation, the nature of dark matter and energy, the creation of the universe through the Big Bang, as well as many other astrophysical and cosmological events. Although previously mostly composed of theoretical subfields, gravitational physics has since expanded to include exciting new experimental prospects.

In our review, we have established some of the theoretical motivations behind seeking alternative theories to general relativity, in particular the disputes regarding the existence of dark matter and energy. We have also discussed one of the leading modified theories of gravity, $f(R)$ gravity, and a prominent alternate theory of gravity, Moffat's Scalar-Tensor-Vector gravity. In addition to this, we explored the exciting new prospects of the relatively recent field of gravitational wave astronomy, such as the ability to investigate the properties of black holes and neutron stars and potentially probe into some of the earliest moments of the universe. We also investigated some of the more recent breakthroughs in gravitational waves research, such as the computation of the Hubble constant through gravitational wave events and the announcement of the gravitational wave background through NANOGrav data. We then concluded by briefly assessing some of the leading ideas of quantum gravity: string theory and loop quantum gravity.

Overall, despite general relativity having officially passed its 100 year anniversary [3], gravitational physics is just now hitting new frontiers and possibilities when it comes to exploring the nature of spacetime. With the ability to now utilize data to test long-standing theories and motivate development of new ones, as well as merge the study of gravitational physics with other areas of cosmology, it is clear we are entering a new age of gravity.

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